

whospoke

Speaker-attributed transcription: who spoke what, and when, in noisy Hindi-English conversations.

4 / 4 milestones complete

THE PIPELINE, IN ORDER

M2 Who speaks when
Speaker embeddings + our own clustering

Darshan

M1 Separate overlaps
Conv-TasNet, only where voices overlap

Vismay

M3 Transcribe
IndicConformer + Hinglish romaniser

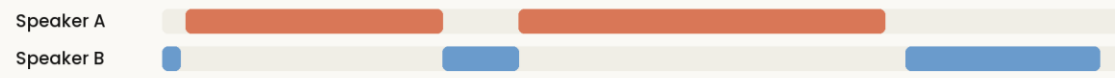
Abhinav

M4 Write the report
Airavata 7B LLM, 4-bit, on a laptop GPU

Shrivaths

TEST CONVERSATION · 2 SPEAKERS · HEAVY OVERLAP · MARKET NOISE

00:44



WHAT WHOSPOKE WROTE, FROM THIS AUDIO ALONE

Speaker A sir mere koi galti hui mere haath se wo mera aai D card gum ho gaya hai sir

Speaker B haan aur baki sun

Speaker A haan chalo exam paas aa rahe hain to aai D card chahiye aadhar card ki photo kapi aur kya kya lana hai sir ek khud ka photo ek aadhar card ki aur

Speaker B apne camera si si ti V camera check karva lijie

Test conversation, 18 s: real Hindi speech and real market noise. Timeline and transcript are whospoke's own output.



00:28–00:35, both talking at once, pulled apart by Conv-TasNet (M1).

All four milestones work on unseen test audio

Held-out test set: 72 conversations (83 min) built from real conversational Hindi (IndicVoices) and real village and market noise. Test speakers never used for tuning.

Where the errors come from

cpWER: a word is wrong if misheard or given to the wrong speaker

ASR alone: clean voices, true speaker timeline



+ noise and overlapping speech (true timeline)



+ our diarization = full pipeline (Order B)



Proposal's order (A): separate first, then diarize · DER 26.2%



Ours: 11% fewer errors, better in 50 of 72 conversations

M1 SEPARATION

+9.6 dB

SI-SDR gain on
overlapping speech (Conv-
TasNet)

M2 DIARIZATION

20.3%

DER, our clustering;
pyannote 3.1: 18.4%

M3 TRANSCRIPTION

21%

WER on real-world Hindi
(Vaani); IndicWav2Vec:
38%

M4 LLM REPORT

~1 min

per minute of audio:
Airavata 7B, 4-bit, 6 GB
laptop GPU

The report, and where we go next

REAL REPORT OUTPUT

Key point

Speaker A lost his ID card

Line 13 · Speaker_A · 00:51

to ab kab tak kitne baje tak office
khula hai sir apna

“The office is open until what time?”

Summary, keywords and an English line for every line: Airavata 7B, 4-bit, run locally with llama.cpp.

What we learned

It never touches who or when. Its JSON output has no speaker or time fields; both are copied from diarization.

Its “repair” does not help. +0.9 points of cpWER on 63 test conversations, so the ASR transcript stays the record.

Errors cascade. Keywords really said: 99% from the true transcript, 60% from real ASR.

Where we go next

Who-spoke-when. About 20 of the 49.5 points, mostly speaker confusion. Next: fine-tune the speaker embeddings on Hindi conversations.

Action items. Empty in all 252 reports from the 7B model. Next: a larger LLM, one setting away.

Separation. Conv-TasNet was trained on English mixtures (Libri2Mix). Next: Indian voices and noise.

One command, recording to report: `python -m whospoke run recording.wav --postprocess`